

Mohammad Siddiquir Rahman

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Orlando, FL - 32826, USA

WORK AND RESEARCH EXPERIENCE

- The University of Central Florida** August 2023 - Current
Graduate Research Assistant Orlando, FL
 - Working as a graduate research assistant in a Department of Energy-funded project
 - Working to implement weather outage modeling to predict power outages in Florida
 - Taken courses related to machine learning, data analysis
- The University of Alabama** August 2021 - August 2023
Graduate Teaching Assistant Tuscaloosa, AL
 - Taken courses related to GIS, Remote Sensing, and Environmental data analysis
 - Developed solid skills in coding, designing research, and drafting manuscripts
 - Was TA in 'Intro to GIS' course and also taught lab section of this course.

EDUCATION

- The University of Central Florida** August 2023 - Spring 2027
PhD in Water Resources Engineering Orlando, FL
- The University of Alabama** August 2021 - August 2023
Master's in Geography Tuscaloosa, AL
- Begum Rokeya University** December 2019
Bachelor in Disaster Management Rangpur, Bangladesh

SKILLS

- Programming Languages:** Python, R, JavaScript, MATLAB
- Database Systems:** MySQL, PostgreSQL, Microsoft SQL Server
- Data Science & Machine Learning:** Apache Spark, Dask, OSMnx, Geopandas, Xarray, Scikit-learn, TensorFlow, PyTorch, Keras, PyCUDA
- DevOps & Version Control:** Git, GitHub, GitLab, SourceTree
- Specialized Area:** Hydrologic Modeling, Geospatial Data Analysis (ArcGIS Pro, ArcPy, ArcGIS Online, Python), Big Data Analytics, High Performance Computing, Parallel Computing, Machine Learning
- Mathematical & Statistical Tools:** Spatial Statistics, Probability Frequency Analysis, Time Series Analysis, Hypothesis Testing
- Research Skills:** Literature Review, Research Design Methodology, Data Collection Management, Data Analysis Interpretation, Communication Dissemination

FUNDED PROJECTS

- Clean, Affordable, and Resilient Energy Systems (CARES)** August 2023 – July 2025
Tools: HPC, Python, R, ArcGIS, PyTorch, TensorFlow 
 - Developed weather power outage modeling for predicting outages based on the extremeness of weather.
 - Assessed the risk and reliability of the Grid to identify vulnerable communities.
 - Studied characteristics of compound extreme weather events and power outages in Florida.
- The Megalopolitan Coastal Transformation Hub (MACH)** July 2025 – Present
Tools: HPC, Python, R, ArcGIS, PyTorch, TensorFlow, PyCUDA
 - Part of the MACH Power System Modeling team.
 - Working on wide area power outage modeling to project and hindcast future and past power outages risk.
 - Use probabilistic framework to model sensitivity of power outages to changes of extreme weather events.

- [C.1] **Rahman, M. S.**, Wahl, T., Nagaraj, M., Rodriguez Enriquez, A. (2024). **Compound Weather Extremes and Power Outages in Florida**. In *AGU Fall Meeting*, 2024, Washington, D.C.
- [J.1] **Rahman, M. S.**, Wahl, T., Nagaraj, M., Enríquez, A. R. (Year). **Characteristics of Power Outages from Compound Weather Extremes in Florida**. *Environmental Research: Climate* [In Press]
DOI:10.1088/2752-5295/adf252
- [J.2] Stevens, K. A., Aksha, S. K., **Rahman, M. S.**, Trader, E., Longenecker, H., Entress, R., Wahl, T., Emrich, C. T., Davis, K. O. (2025). **Shedding light on inequities of power outages through data transparency**. *Progress in Energy*, 7(3), 033002. DOI:10.1088/2516-1083/addc30
- [J.3] Belligoni, S., Trader, E., Li, M., **Rahman, M. S.**, Ali, J., Enriquez, A. R., Nagaraj, M., Aksha, S. K., Stevens, K. A., Wahl, T., Emrich, C. T., Alam, N., Ganguly, A. R., Bukvic, A., Dinh, C., Pham, K. D., Nguyen, T. D. (2025). **Transdisciplinary research promoting clean and resilient energy systems for socially vulnerable communities: A review**. *Renewable and Sustainable Energy Reviews*, 213, 115434. DOI:10.1016/j.rser.2025.115434
- [J.4] **Rahman, M. S.**, Senkbeil, J. C., Keellings, D. J. (2023). **Spatial and Temporal Variability of Extreme Precipitation Events in the Southeastern United States**. *Atmosphere*, 14(8), 1301. DOI:10.3390/atmos14081301
- [J.5] Musabbir, M., Islam, A. R. M. T., **Rahman, M. S.**, Pal, S. C., Alam, E., Mallick, J. (2023). **Spatiotemporal variability of rainfall erosivity and its teleconnection with atmospheric circulation in monsoon-driven climate region**. *CATENA*, 222, 106762. DOI:10.1016/j.catena.2022.106762
- [J.6] Azad, M. A. K., Islam, A. R. M. T., Ayen, K., **Rahman, M. S.**, Shahid, S., Mallick, J. (2022). **Changes in monsoon precipitation patterns over Bangladesh and its teleconnections with global climate**. *Theoretical and Applied Climatology*, 149(1–2), 651–669. DOI:10.1007/s00704-022-04021-w
- [J.7] Azad, M. A. K., Islam, A. R. M. T., **Rahman, M. S.**, Ayen, K. (2021). **Development of novel hybrid machine learning models for monthly thunderstorm frequency prediction over Bangladesh**. *Natural Hazards*, 106(3), 2633–2659. DOI:10.1007/s11069-021-04532-5
- [J.8] **Rahman, M. S.**, Azad, M. A. K., Hasanuzzaman, M., Salam, R., Islam, A. R. M. T., Abdullah, K. P., Baten, S. M. A., Akter, T. T., Hasan, S. M. D. (2021). **How air quality and COVID-19 transmission change under different lockdown scenarios? A case from Dhaka city, Bangladesh**. *Science of The Total Environment*, 754, 143161. DOI:10.1016/j.scitotenv.2020.143161
- [J.9] Islam, A. R. M., Ahmed, I., **Rahman, M. S.** (2020). **Trends in cooling and heating degree-days overtimes in Bangladesh? An investigation of the possible causes of changes**. *Natural Hazards*, 104(2), 1397–1417. DOI:10.1007/s11069-020-04213-7
- [J.10] Islam, A. R. M. T., **Rahman, M. S.**, Khatun, R., Hu, Z. (2020). **Spatiotemporal trends in the frequency of daily rainfall in Bangladesh during 1975–2017**. *Theoretical and Applied Climatology*, 141(3–4), 1523–1539. DOI:10.1007/s00704-020-03244-x
- [J.11] **Rahman, M. S.**, Islam, A. R. M. T. (2019). **Are precipitation concentration and intensity changing in Bangladesh overtimes? Analysis of the possible causes of changes in precipitation systems**. *Science of the Total Environment*, 683, 127008. DOI:10.1016/j.scitotenv.2019.06.529

CONFERENCES

- AGU Annual Meeting 2024, Washington, D.C. December 9-13, 2024
- ASEMFL 2024 Annual Meeting, Orlando, FL November 1-2 2024
- 49th Annual Natural Hazards Research and Applications Workshop, Boulder, CO July 14-17, 2024
- 2023 AAG Annual Meeting, Denver, CO March 23-27, 2023

SUMMER SCHOOL EXPERIENCE

- NSF I-GUIDE Summer School: Leveraging AI for Environmental Sustainability (funded) August 5-9, 2024
Tools: Python, PyTorch, TensorFlow, GitHub 
 - Developed a machine learning model using PyTorch to predict the efficacy of biochar in removing pharmaceutical waste from water sources.
 - Collaborated in a team to analyze results and presented key findings to researchers from NCAR and other universities.
- MACH Summer School 2025 August 5-8, 2025
networking, knowledge sharing, communication 
 - Selected to participate in an intensive program featuring expert lectures, collaborative research discussions, and a technical field trip with fellow MACH members.

HONORS AND AWARDS

- **Graduate Dean's Fellowship**

University of Central Florida

August 2023



- Awarded to superior graduate students newly enrolling in eligible programs based on academic excellence.

- **Faculty Cluster Initiative Fellowship**

University of Central Florida

August 2025

- Selected for fellowship supporting multidisciplinary research focused on solving complex, real-world problems.

PROFESSIONAL MEMBERSHIPS

- **American Geophysical Union**

August 2024 - Present

ADDITIONAL INFORMATION

Research Interests: Hydro-climatic Extremes, Compound Extremes, Power Outage Modeling, Machine learning

General Interests: Data Analysis and Visualization, Badminton, Following tech updates

REFERENCES

1. **Dr. Thomas Wahl**

Associate Professor, Water Resource Engineering and UCF Coastal

University of Central Florida

Email: t.wahl@ucf.edu

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Relationship: Ph.D. Advisor

2. **Dr. Jason Senkbeil**

Professor, Department of Geography

The University of Alabama

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Relationship: Master's Thesis Supervisor